devices, bionic organs and nano-robots, which will monitor our health and defend us from infections, illnesses and damage. Yet these devices will have to be online 24/7, both in order to be updated with the latest medical news, and in order to protect them from the new plagues of cyberspace. Just as my home computer is constantly attacked by viruses, worms and Trojan horses, so will be my pacemaker, my hearing aid and my nanotech immune system. If I don't update my body's anti-virus program regularly, I will wake up one day to discover that the millions of nano-robots coursing through my veins are now controlled by a North Korean hacker.

The new technologies of the twenty-first century may thus reverse the humanist revolution, stripping humans of their authority, and empowering non-human algorithms instead. If you are horrified by this direction, don't blame the computer geeks. The responsibility actually lies with the biologists. It is crucial to realise that this entire trend is fuelled by biological insights more than by computer science. It is the life sciences that have concluded that organisms are algorithms. If this is not the case – if organisms function in an inherently different way to algorithms – then computers may work wonders in other fields, but they will not be able to understand us and direct our life, and they will certainly be incapable of merging with us. Yet once biologists concluded that organisms are algorithms, they dismantled the wall between the organic and inorganic, turned the computer revolution from a purely mechanical affair into a biological cataclysm, and shifted authority from individual humans to networked algorithms.

Some people are indeed horrified by this development, but the fact is that millions willingly embrace it. Already today many of us give up our privacy and our individuality, record our every action, conduct our lives online and become hysterical if connection to the net is interrupted even for a few minutes. The shifting of authority from humans to algorithms is happening all around us, not as a result of some momentous governmental decision, but due to a flood of mundane choices.

The result will not be an Orwellian police state. We always prepare ourselves for the previous enemy, even when we face an